

CREDIT BUREAU SYSTEM & SCORING MODEL REVIEW REPORT

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The system demonstrates strong progress and is largely aligned with the Bank of Ghana (BOG) Input File Formats (IFF) and credit reporting framework.

The credit scoring model is well developed, with model validation metrics already implemented and performing at acceptable levels.

However, a few critical implementation gaps remain, particularly in:

- Search parameter structuring
- Data upload (IFF integration)
- PDF report formatting
- Full integration of NDIA & Amount in Arrears into scoring logic

Addressing these will ensure:

- Full regulatory compliance
- Accurate borrower identification
- Robust and explainable credit scoring
- Market readiness

2. System Implementation Status

#	Component	Requirement	Status	Remarks
1	Consumer Credit Reports	Separate structure	Completed	Well aligned and structured
2	Business Credit Reports	Separate structure	Completed	Proper segregation achieved
3	Search Portal Structure	Consumer vs Business separation	Partial	Needs refinement and enforcement
4	Consumer Search Parameters	Ghana Card-based structured fields	Not Completed	Critical for identity matching
5	Business Search Parameters	Legal entity identifiers	Not Completed	Required for accurate mapping
6	Data Upload Module (IFFs)	Upload interfaces for all 6 IFFs	Not Completed	Core regulatory requirement
7	PDF Credit Report Output	Structured, readable format	Not Completed	Currently distorted

3. Search Parameter Framework (Critical Gap)

3.1 Consumer Search Parameters

Field	Requirement	Purpose
Ghana Card Number	Mandatory (Primary ID)	Unique identification
First Name	Mandatory	Identity matching
Middle Name	Optional	Additional matching
Last Name	Mandatory	Identity validation
Date of Birth	Mandatory	Identity confirmation
Mobile Number	Mandatory	Secondary identifier
Gender	Mandatory	Profile completeness
Email	Optional	Contact reference
Other ID Types	Optional	Legacy loan mapping
Reason for Request	Mandatory	Audit/compliance
Amount Requested	Optional	Lending context
Report Type	Mandatory	Output classification

3.2 Business Search Parameters

Field	Requirement	Purpose
Business Name	Mandatory	Entity identification
Registration Number	Mandatory (Primary ID)	Legal identity
Registration Date	Mandatory	Verification
TIN Number	Optional	Tax mapping
Purpose of Inquiry	Mandatory	Compliance
Amount Requested	Mandatory	Lending context

4. Data Upload & IFF Integration (Critical Requirement)

4.1 Required Upload Modules

Data Type	Description	Status
Consumer Credit Data	Individual facilities	Not Implemented
Business Credit Data	Commercial facilities	Not Implemented
Consumer Judgment Data	Court records (individuals)	Not Implemented
Business Judgment Data	Court records (companies)	Not Implemented
Consumer Dishonoured Data	Returned instruments	Not Implemented
Business Dishonoured Data	Returned instruments	Not Implemented

4.2 Functional Requirements

Requirement	Description
File Validation	Enforce IFF structure
Business Rules	NDIA & Ammount in Arrears & arrears logic
Error Reporting	Clear rejection feedback
Data Segregation	Consumer vs Business
Audit Logs	Track uploads

5. Credit Report Output (PDF) – Structural Issue

Issue	Impact	Recommendation
Distorted layout	Poor readability	Align all sections properly
Inconsistent formatting	User confusion	Standardize tables & headers
Missing structure flow	Weak usability	Follow logical report sections

6. Credit Scoring Model Review

6.1 Scoring Factors Assessment

Factor	Status	Recommendation
Payment History	Good	Must be NDIA & Amount In Arrears driven
Credit Utilization	Good	Enhance with exposure ratios
Credit History Length	Good	Retain
Credit Inquiries	Good	Define thresholds
Account Mix	Good	Retain
NDIA (Arrears Severity)	Missing	Must be integrated

7. NDIA & Amount in Arrears-Based Risk Framework (Mandatory)

NDIA Range	Risk Level	Score Impact
0	Performing	Positive
1-30	Early delinquency	Slight negative
31-90	Moderate risk	Medium negative
91-180	High risk	Strong negative
180+	Default	Severe negative

8. Score Band Structure

Score Range	Rating	Risk Level	Lending Guidance
750-850	Excellent	Very Low Risk	Approve
670-749	Good	Low Risk	Approve
580-669	Fair	Medium Risk	Conditional
450-579	Poor	High Risk	High caution

300-449	Very Poor	Very High Risk	Decline
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Recommendation

- Minimum 500-1,000 records per segment
- Balanced dataset for:
 - Good vs Bad borrowers
 - Realistic default distribution

9. Model Validation Metrics (Confirmed & Strong)

The system already includes model validation metrics on each credit report.

Validation Results

Metric	Description	Result	Assessment
GINI Coefficient	Predictive power	0.62	Strong
KS Statistic	Risk separation	0.48	Strong
Rank Ordering	Score ranking	Validated	Acceptable
Stress Testing	Model robustness	Passed	Acceptable
Probability of Default	Risk estimation	Per account	Implemented

Interpretation

- The model demonstrates good discriminatory power
- Risk separation between good and bad borrowers is adequate
- The scoring model is statistically sound
- Suitable for operational deployment, subject to final refinements

10. Score Band Performance (Observation)

Issue	Impact
Small dataset	Limits reliability
Uneven distribution	Affects calibration

Recommendation

- Increase dataset size
- Improve segmentation across score bands

11. Reason Codes Review

Assessment: **Well structured**

Enhancements Required

Code	Description
HIGH_NDIA_90_PLUS	Accounts >90 days overdue
MULTIPLE_DELINQUENCIES	Repeated late payments
HIGH_ARREARS_AMOUNT	Significant overdue balances

12. Next Phase Requirements

Area	Requirement
Matching Algorithm	Identity matching logic (Ghana Card + fuzzy logic)
Score Algorithm	Full formula breakdown
Data Set	Training data for validation
Model Calibration	Local market tuning

13. Conclusion

The system and scoring model are well advanced and near deployment readiness.

Strengths include:

- Strong system structure
- Functional credit reporting modules
- Robust scoring model with validated metrics

However, final implementation of:

- Search parameters
- IFF upload modules
- NDIA & Amount in Arrears driven scoring logic
- Report formatting

is required before final certification and go-live approval.

14. Final Next Steps

1. Complete outstanding system enhancements
2. Confirm readiness for final validation review
3. Proceed to final system certification

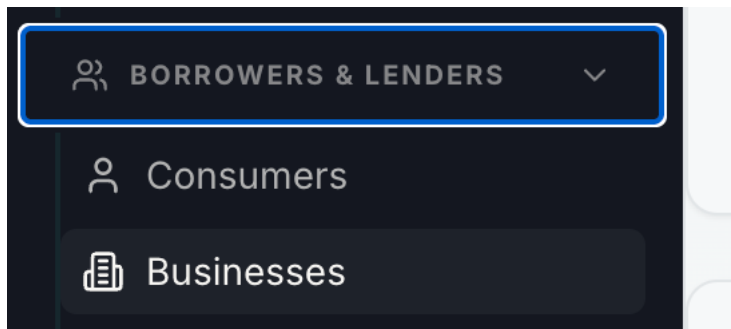
Final Note

Upon completion of the above:

- The system will meet Ghana regulatory standards

- The scoring model will be fully compliant and reliable
- The platform will be ready for market deployment and scaling

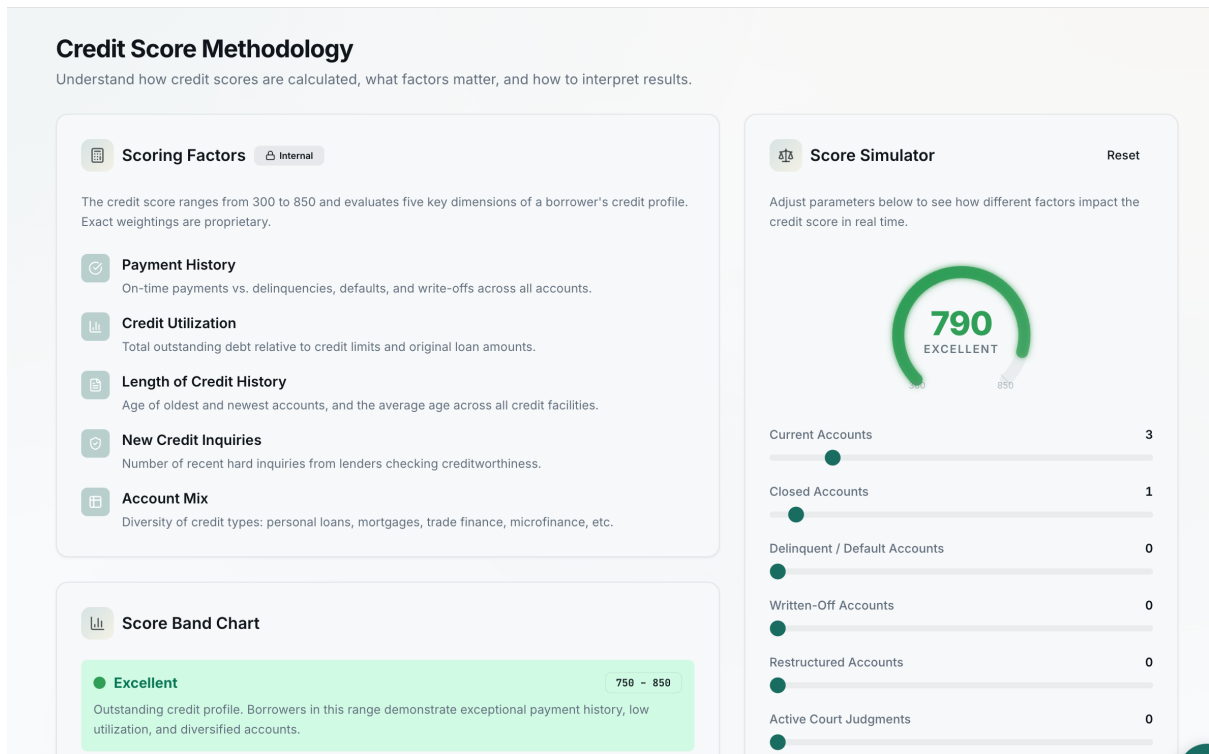
Screenshots from the system as at today's review.



This is completed.

A screenshot of a web form titled 'Generate Credit Report'. The form has a dark green header with the title and a subtitle 'Search for a borrower and generate a comprehensive D&B-style credit report'. Below the header, there is a section '1. Find Borrower'. This section contains a search input field with a magnifying glass icon and placeholder text 'Search by name, national ID, TIN, or passport...'. To the right of the input field is a dropdown menu showing 'Ghana'. Below the input field is a magnifying glass icon and the text 'Search for a borrower above to generate a credit report'. At the bottom, there is a note: 'You can search by name, national ID, TIN, passport number, or company name'.

Search parameters – should follow guide provided above



Credit score methodology.

Batch Upload

Upload credit account data in bulk via JSON or CSV. Invalid records are rejected with error details. [CSV Template](#)

Required Borrower Fields (all formats)

`borrowerName` `dateOfBirth` `address` `nationalId` `phoneNumber` `reportingDate`

These fields are required in addition to account fields (borrowerId, lenderInstitution, accountNumber, etc.) for accurate borrower matching and credit report generation.

CSV Upload

Upload CSV file
Comma-separated values (.csv)

Or paste CSV directly:

Paste CSV data with headers...

[Validate & Preview](#) [Submit CSV Batch](#)

CSV Sample Format

```
borrowerId,borrowerName,dateOfBirth,address,nationalId,phoneNumber,reportingDate
BORROWER_ID_1,Kwame Mensah,1985-03-15,"12 Independence Ave",0901234567,2023-01-01
BORROWER_ID_2,Abena Osei,1990-07-22,"45 Ring Road",0901234567,2023-01-01
```

[Use Sample Data](#)

Data submission Portal – Follow guide provided above.